📅 Training Day 5 Report

Date: 21 June 2025
Module Name: OSI Model – Detailed Explanation and Real-World Relevance
Mode: Online

🔧 Main Focus of the Module

On Day 5, the training session was dedicated to understanding the **OSI (Open Systems Interconnection) Model** in detail. This conceptual model is fundamental to computer networks and cybersecurity, as it helps explain how data moves through a network. Each layer of the OSI model was discussed in-depth along with **protocols, real-world examples, and attacks relevant to each layer**.

📚 Topics Covered

🖧 What is the OSI Model?

- The OSI Model is a 7-layer framework that standardizes the functions of a telecommunication or computing system.
- Each layer serves the one above it and is served by the one below it.
- Helps in **troubleshooting**, **protocol design**, and **understanding security flaws**.

OSI Model – Layer-by-Layer Breakdown

Layer	Name	Description	Protocols/Examples	Common Attacks
7	**Application**	Interfaces with end-user software	HTTP, FTP, DNS, SMTP	Phishing, DNS Poisoning
6	**Presentation**	Data formatting, encryption, compression	SSL/TLS, JPEG, ASCII	SSL Stripping, Code Injection
5	**Session**	Controls sessions between computers	NetBIOS, RPC, PPTP	Session Hijacking
4	**Transport**	Reliable delivery, flow control	TCP, UDP	TCP Flood, UDP Flood
3	**Network**	Logical addressing, routing	IP, ICMP, IGMP	IP Spoofing, ICMP Flood
2	**Data Link**	MAC addressing, switches	Ethernet, ARP	ARP Spoofing

Layer	Name	Description	Protocols/Examples	Common Attacks
1	**Physical**	Transmission of raw bits	Cables, Hubs, NICs	Wiretapping, Jamming

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📌 Summary of Each Layer:

1. **Application Layer (Layer 7)**
   - Closest to the user
   - Provides network services to applications
   - E.g., Web browsers use HTTP
2. **Presentation Layer (Layer 6)**
   - Translates, encrypts, compresses data
   - Ensures proper syntax and format
3. **Session Layer (Layer 5)**
   - Manages sessions between devices
   - Responsible for establishing, maintaining, and terminating connections
4. **Transport Layer (Layer 4)**
   - Ensures error-free transmission
   - Provides segmentation, flow control
   - TCP (reliable), UDP (unreliable)
5. **Network Layer (Layer 3)**
   - Handles logical addressing and routing
   - IP addressing and path selection
6. **Data Link Layer (Layer 2)**
   - Responsible for node-to-node data transfer
   - MAC addressing and framing
   - Divided into: Logical Link Control (LLC) & Media Access Control (MAC)
7. **Physical Layer (Layer 1)**
   - Actual transmission of raw data bits over physical medium
   - Deals with voltage, cable type, pin layout

? Hands-On / Real-World Examples

- **Captured packets using Wireshark** and observed how each layer is represented in actual data flow.
- Identified TCP/UDP headers (Layer 4), IP packets (Layer 3), and MAC addresses (Layer 2).
- Discussed how an attack like **ARP spoofing** exploits Layer 2 and **DNS spoofing** hits Layer 7.

🧠 Key Takeaways

- The OSI Model is essential for understanding how data is structured and transmitted.

- Many cyberattacks are **layer-specific**, and knowing the model helps in **detecting and defending** against them.
- Tools like **Wireshark**, **Nmap**, and **Metasploit** function across multiple lay